

LESSON 138 (0.5) Ground (1.0) Local Dual Night (5) Landings

Lesson Objective

The purpose of this lesson is to introduce the applicant to the operational aspects of night flight. Emphasis will be placed on the additional planning necessary when operating at night. The flight will consist of both traffic pattern and practice area operations.

A minimum of 10 takeoffs and landings to a FULL STOP must be completed by lesson 139.

Pre-Brief

- Review Maneuvers and Procedures to be Performed in Flight.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Discuss previous solo if lesson was closed out by dispatch.
- Student questions.

References

- Airplane Flying Handbook. (FAA-H-8083-3C)
 - Chapter 11: Night Operations.
 - Chapter 18: Emergency Procedures.
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
 - Chapter 17: Aeromedical Factors.

Instructor Guidance

- Ensure the student has the correct materials for the flight:
- Red and green flashlight when inside the flight deck and white flashlight for preflight.
- When on the ramp, emphasize that disorientation and loss of night vision can be caused by strobe lights. If able, use the beacon only when taxiing near other aircraft. Ensure navigation lights are on before engine start.
- Areas of emphasis at night — maintaining visual references is significantly more difficult. Make full use of the flight instruments:
- Bug the heading on the HSI.
- Maintain a constant scan of the attitude indicator, altimeter, VSI, and turn coordinator.
- Pick a well-lit area or a bright light in the distance to use as a visual reference for alignment.
- Ensure the student is being introduced to night illusions throughout the flight:
- Fly over the shoreline to demonstrate the lack of visual references over the ocean.
- Point out potential false horizons as they occur.
- Verify the student is using off-center viewing when scanning for traffic.
- Acknowledge that the student may tend to round out too high during night landings — actively assist with height perception during the flare and brief this tendency before the first approach.
- Use the landing light as needed to aid visibility during taxi.

Lesson Content**Checklist Procedures**

- Have the student run through all checklists.
- Incorporate flows for memory items.
 - Student should be able to perform all briefings.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.
 - Verify student uses a bright white flashlight to check all fluids.

Operation of the Communications and Navigation Units

- Ensure that the student can proficiently use COMS and NAV units.

Engine Starting

- Correctly determine and use appropriate start procedure.
- Incorporate the use of flows.

Radio Communication Procedures and ATC Phraseology

- The student should be able to conduct radio calls at towered and non-towered airports.
- The student should be able to conduct radio calls in the practice area.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!
- At night, use peripheral vision to scan for traffic.
 - Avoid staring at stationary lights for too long.
 - If traffic is spotted, use off-center viewing to keep visual contact with it.

Taxiing

- Emphasize that lack of visual references may cause the student to taxi faster than normal.
 - Have student monitor ground speed as necessary.
- The student should be able to taxi on the centerline at appropriate speeds on the ramp and taxiways without instructor guidance.
- Ensure the correct use of wind correction throughout taxi.

Normal and Crosswind Takeoff

- Takeoff procedures at night are the same as during the day.
 - If able, avoid rolling takeoffs at night.
 - Intersection departures are NOT ALLOWED.
- Areas of emphasis:
 - Clear the area thoroughly. It is harder to identify a clear runway and approach and departure at night.

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- Ensure the magnetic compass and HSI match with the intended runway centerline.
- Verify line up check is complete.
 - Check that centerline is maintained using the lights to verify when able.
 - Confirm that runway edge lights are correctly spaced at both sides from centerline.
- Perception of runway width, airplane speed, and flight attitude will vary at night.
 - Check flight instruments frequently.
- The ensure airplane is climbing, check the airspeed indicator, VSI, and altimeter;
 - Darkness makes it hard to judge rate of climb.
- Ensure turns are made at safe altitudes.

Maneuvering During Slow Flight

- Student says, student does.
 - Verify correct procedure.
- If able, use a light in the distance or well-lit area to reference for the maneuver.
- Ensure student is scanning instruments frequently to maintain coordination and verify they are staying within the standards of the maneuver.

Stalls

- Student says, student does.
 - Verify correct procedure.
- If able, use a light in the distance or well-lit area to reference for the maneuver.
- Ensure student is scanning instruments frequently to maintain coordination and verify they are staying within the standards of the maneuver.

Traffic Patterns

- Stable approaches and landings should be made in the same manner as during the day.
 - If unable to see obstructions to your approach path, try to maintain a slightly higher approach.
- Areas of emphasis at night:
 - Be cautious of featureless terrain/black hole approach.
 - Clear the area thoroughly.
 - Use flight instruments more often:
 - Distance, height, etc. may be deceptive.
 - Maintain specified airspeed on each leg and watch the VSI to keep the approach under control.
- At non-towered airports with pilot lighting, allow the student to turn the lights on using the mic button.

Systems and Equipment Malfunctions (simulated electrical failure)

- Simulate an electrical failure by dimming the PFD.
- Ensure the student can run through the proper checklists and verify them.
- Ensure the student is reducing the electrical load as necessary.

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- Question the student of their decision making.
 - Is this a land “as soon as practical” or land “as soon as possible” emergency?

Go-Arounds or Rejected Landings

- Have your student practice go-around procedures.
- Encourage your student to call for a go-around themselves rather than waiting for you to call it.
- If you do call for a go-around, the student should immediately execute the procedure rather than discussing it first.

Normal and Crosswind Landings

- Student should conduct a normal/crosswind landing.
- Ensure runway heading is verified and final check is completed.
 - Verify the magnetic compass and HSI match with the intended runway centerline.
 - Perception of runway width, airplane speed, and flight attitude will vary at night.
 - Check flight instruments frequently.
 - Use approach lighting systems to maintain desired glide path.
 - Check chart supplement prior to flight to see available lighting.
 - Verify lights are working/usable via NOTAMs.
 - Judgement of height, speed, and sink rate may be impaired.
 - There is a tendency to round out too high.
 - Start the round out when the light starts to reflect off the tire marks on the runway.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post Brief

- Student self-critique.
- Instructor critique.
- Student questions.
- Lesson grading.
- Logbook Completion.
- Schedule and preview next lesson.
- Assign Homework:
 - Review night operations.
 - Review cross-country planning procedures.
 - Review emergencies.
 - Ensure flight publications are up to date.
 - Have student plan a cross-country to be flown in lesson 139.
 - Must be more than 100NM and meet the lesson requirement.

Completion Standards

This lesson is complete when the applicant can:

1. Perform preflight actions at night.
2. Conduct maneuvers at night.
 - a. During power-off stalls and power-on stalls:
 - i. Heading ± 15 degrees and airspeed ± 15 knots.
 - b. During slow flight:
 - i. Maintain altitude ± 150 ft, airspeed $+15$ and -0 knots and heading ± 15 degrees.
3. Maintain directional control during takeoffs and landings.
 - a. Demonstrate all landings within stage two completion standards.
4. Identify and correct for night illusions with instructor guidance.